

Multi-Speaker Separation via Multiplicative Beamforming

Passive direction detection, range selectivity,
and source separation
from a compact microphone array
(Thanks to Bob Zurek)

A4R (Audio for Robotics) | June 2026

The Problem: Multi-Speaker in a Living Room

Scenario (from Robotics PRD):

- Family of 4–5 in a living room
- Robot must identify and lock onto **target speaker**
- Distinguish **near-field** (same room) from **far-field** (other room) audio
- Suppress competing voices and ambient noise
- All with minimal compute on embedded hardware

Requirements:

- **Direction detection** — find where each speaker is
- **Range discrimination** — near vs. far sources
- **Audio separation** — isolate the target speaker's voice
- **Low compute** — run on robot CPU budget
- **Compact array** — fit on robot head

Core Concept: 2-Mic Multiplicative Beamforming

From just **2 microphones**, we derive two spatial primitives that share the same on-axis frequency response:

Cardioid (C)

Delay-and-subtract:

$$m1 - m2 \cdot e^{j\omega\tau}$$

$$\tau = 2r/c \approx 407 \mu s$$

Heart-shaped pickup pattern

Dipole (D)

Pressure-gradient:

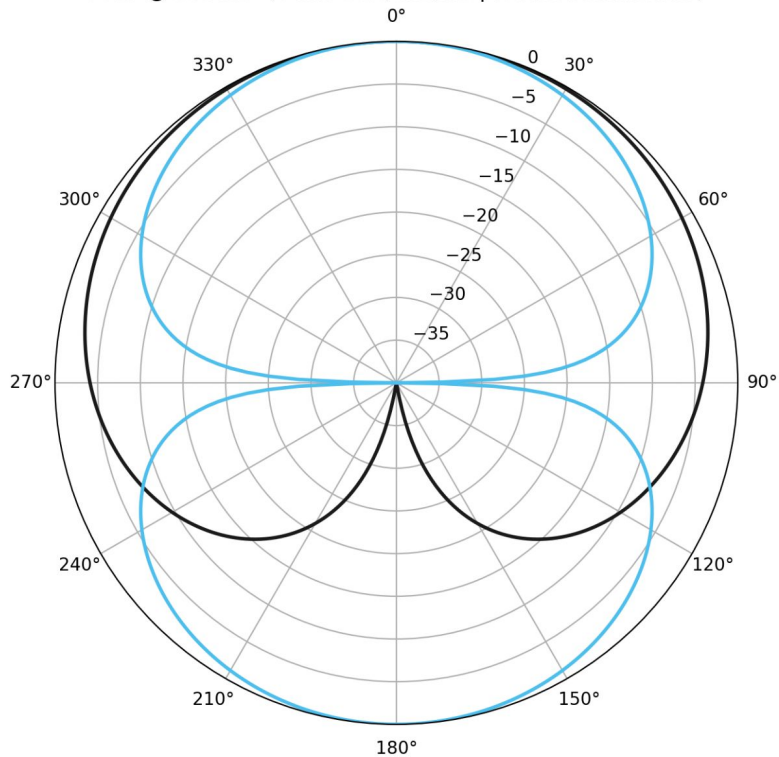
$$m1 - m2$$

Flattened by $2\cos(\omega r/c)$

Figure-8 pickup pattern

Key Insight: Both primitives share the same on-axis response in the differential band, enabling clean multiplicative combinations that modify spatial behavior while preserving spectral shape.

Polar @ 500 Hz (4-mic UCA, m1/m3 product beamformer)



Display instances

- ☒ Cardioid (C)
- ☒ Dipole (D)
- ☐ Dipole² (D²)
- ☐ Dipole³ (D³)
- ☐ 1: C x D
- ☐ 2: C x D²
- ☐ 3: C x D³
- ☐ 4: C²
- ☐ 5: C x D⁵

Array: 4-mic UCA
dia 5.5 in (139.7 mm)
r = 69.85 mm

Pair: m1 (front) / m3 (back)
delay tau = 407 us

Each curve normalized to
its own peak (dB).

Differential band good
below ~1.23 kHz; aliasing
above (watch the slider).

log10(f)  2.600 **500 Hz**

— Cardioid (C) — Dipole (D)

Angular Selectivity: 2 Mics $\approx$ 15-Element Array

Multiplying C and D primitives creates **progressively narrower beams** without additional microphones:

- $C \times D$ — moderately narrow
- $C \times D^3$ — significantly narrower
- $C \times D^5$ — highly directional
- $C^3 \times D^5$ — extremely narrow

Each higher-order product sharpens the beam, enabling precise **speaker direction detection** from just 2 microphones.

vs. Conventional approaches: GCC-PHAT gives only TDOA (time delay) — no spatial filtering. MVDR/MaxSNR optimize noise, not direction. Multiplicative beamforming inherently **separates sources spatially** as a side effect of direction detection.

Why this matters:

2 microphones achieve the angular resolution of a 15-element conventional array

Referenced in academic literature — eliminates need for large, expensive arrays

Range Selectivity: Near-Field vs Far-Field

The same multiplicative products also provide **distance discrimination**:

A free-field source attenuates at 6 dB/oct ($1/r$). Higher-order products attenuate **much faster** — so the ratio between orders is a **monotonic distance proxy**.

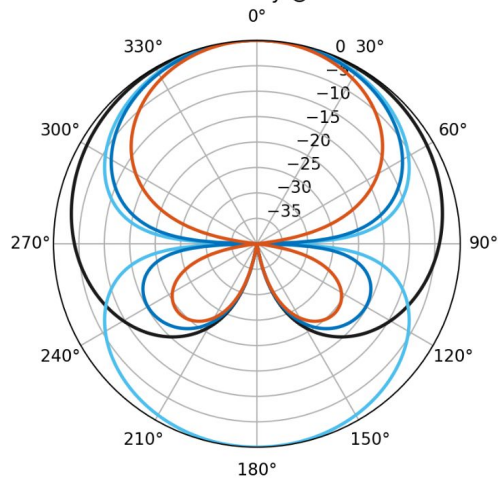
Level-independent distance estimation from a single passive array.

Instance	Order	Slope
C, D	1	-6 dB/oct
$C \times D$	2	-12 dB/oct
$C \times D^3$	4	-24 dB/oct
$C \times D^5$	6	-36 dB/oct

Application to robot: By comparing the ratio of a high-order beam output to the raw pressure, the robot instantly classifies sources as:

- **Near-field** (same room, within ~1–5m) — the target speaker
- **Far-field** (other room, hallway) — to be suppressed

Polar directivity @ 500 Hz



log10(f) 2.699

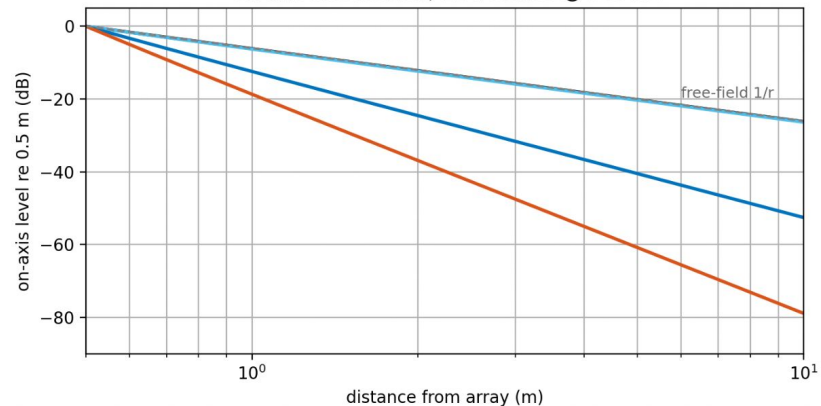
f = 500 Hz

Display instances

- ☒ Cardioid (C)
- ☒ Dipole (D)
- ☐ Dipole^2 (D^2)
- ☐ Dipole^3 (D^3)
- ☒ 1: C x D
- ☒ 2: C x D^2
- ☐ 3: C x D^3
- ☐ 4: C^2
- ☐ 5: C x D^5

— Cardioid (C) — Dipole (D) — 1: C x D — 2: C x D^2

On-axis attenuation, 1 Pa source @ 500 Hz



Array: 4-mic UCA, dia 5.5 in
(139.7 mm, r = 69.85 mm)
Pair m1(front)/m3(back), tau=407us

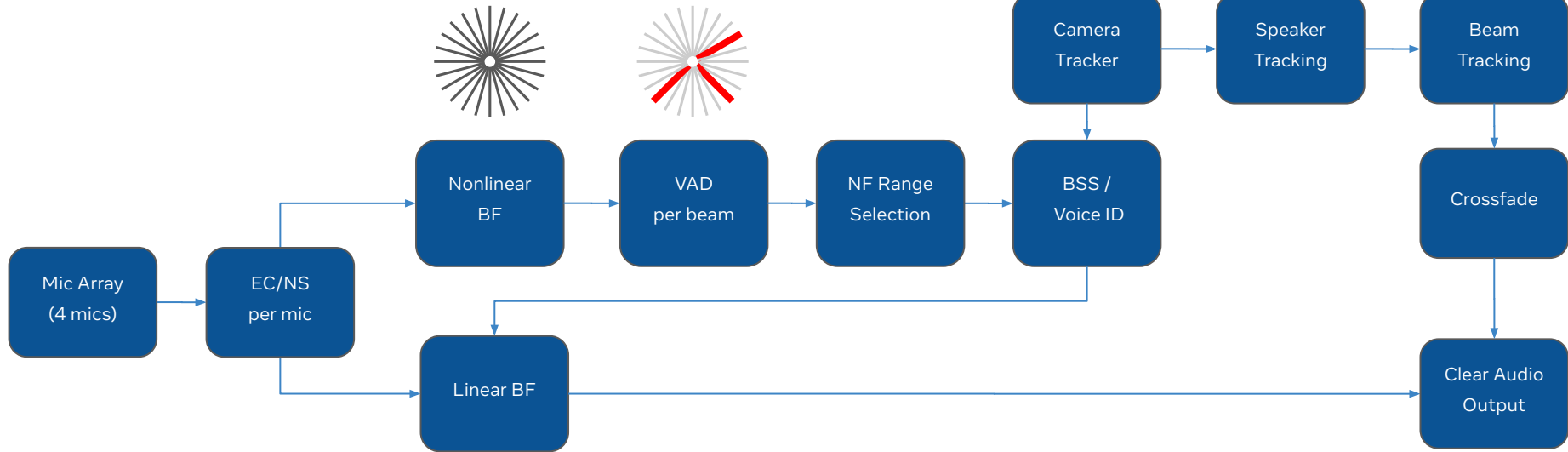
POLAR: far-field directivity,
each curve norm. to its peak.

RANGE: 1 Pa point source on axis,
output vs distance, each curve
norm. to its 0.5 m value.
Product order N -> ~N*6 dB/oct.

Differential band good below
~1.23 kHz; aliasing above.

Direction Path: Direction detection via multiplicative beams

Camera fusion



Audio Path: High-quality audio extraction on beam of interest

Notes:

1. **Nonlinear BF:** Multiplicative beamformer identifies active speaker directions in fine angular granularity
2. **VAD:** VAD filters out non-speech sources (HVAC, appliances)
3. **Linear BF:** MVDR beamformer extracts high-fidelity audio from target beam
4. **BSS:** Blind Source Separation isolates individual speakers within the beam
5. **Voice ID:** Tags individual speakers

Dual-Path Architecture

Two specialized processing paths run in parallel — each optimized for its task:

Direction Path (Control Signal)

Method: Nonlinear / Multiplicative BF

Purpose: Find speaker directions + range

Compute: Very low MIPS

Quality: Spatial info only (not audio)

Rate: 16 kHz sufficient

Narrow beams scan for active sources;
output steers the audio path

Audio Path (Extraction)

Method: Linear / Conventional BF

Purpose: Extract clean target audio

Compute: Moderate MIPS

Quality: High-fidelity audio output

Rate: 16 kHz (or higher for quality)

Steered by direction path; optimizes SNR
for the selected speaker

Benefit: Direction detection runs at negligible cost, freeing compute budget for high-quality audio extraction only where needed.

Computational Efficiency: Built for Embedded

Multiplicative beamforming is **computationally trivial** — multiplication of two already-computed differential signals. No matrix inversions, no iterative solvers.

Comparison with conventional approaches:

Approach	Operations	MIPS	Mics needed
MVDR / MaxSNR	Matrix inversion, covariance est.	High	4–16+
GCC-PHAT	FFT, cross-correlation, peak search	Moderate	2+ (DOA only)
Multiplicative BF	Subtract + multiply	Very low	2 (full separation)

16 kHz sample rate sufficient for both direction and audio paths. 8 kHz potentially viable for direction-only (speech position info largely below 1–1.5 kHz).

Takeaway

Multiplicative beamforming gives us direction + range + separation — all from a compact, passive microphone array with minimal compute.

Angular Selectivity

Find each speaker
direction with 2-mic =
15-element resolution

Range Selectivity

Distinguish near-field
user from far-field
sources

Source Separation

Isolate target speaker via
spatial filtering + BSS

Next Steps:

- Physical demo with nonlinear beamforming on 4-mic array
- Validate angular resolution and range discrimination in living room scenario
- Integrate full pipeline (VAD → direction → extraction → BSS) on robot platform